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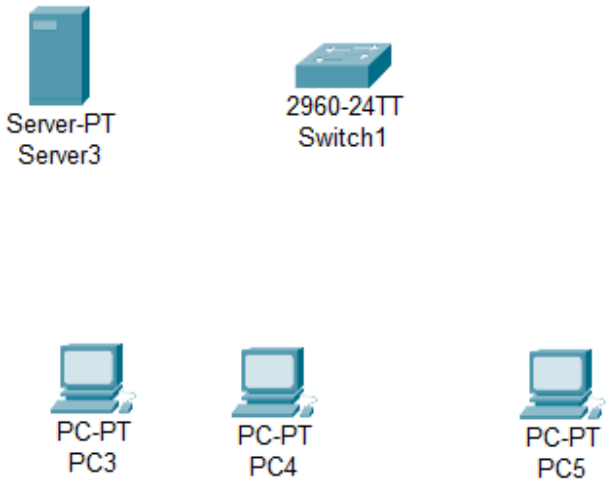
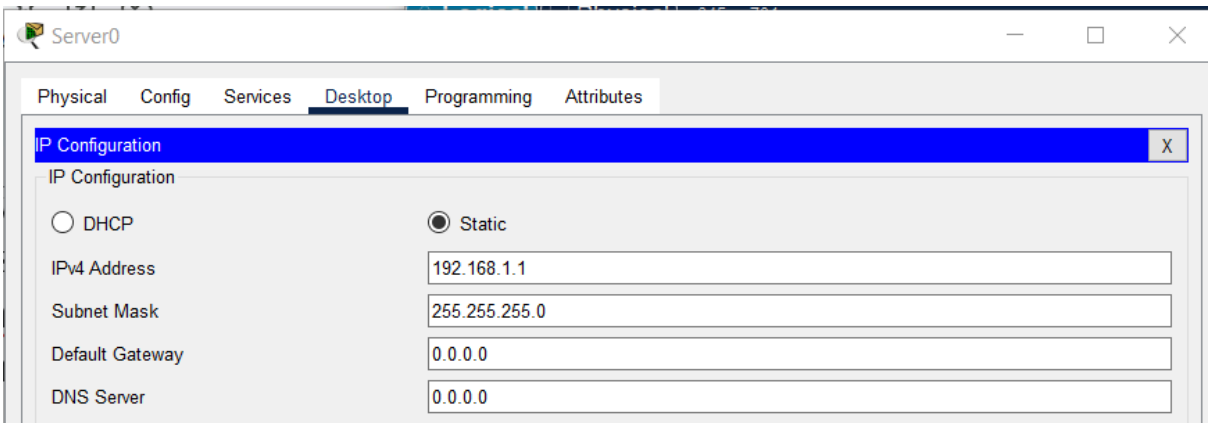
In this document, we are going to see a configuring DHCP server in Cisco Packet Tracer, see the topology and how it works.

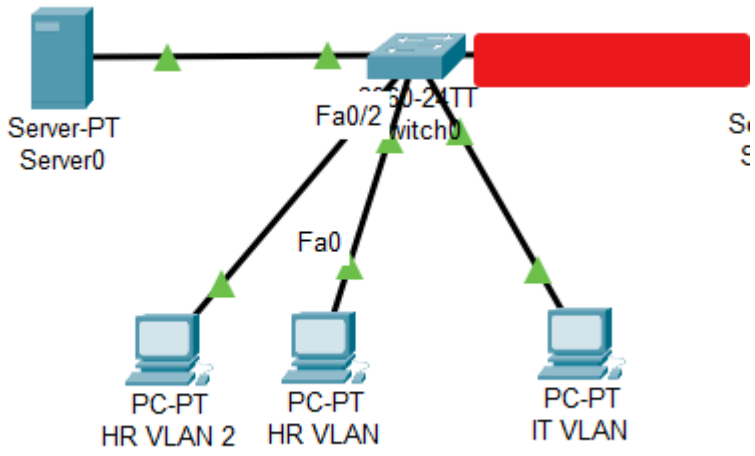
Here we can see what kind of topology we are using,we see one server, one switch and three PCs, two with the same VLAN, with the assigned name HR, and the other with name IT.



Now, we have to assign an IP number to the server, following the next image, we see that we enter in the desktop and assign an IP that was given to us, that's the first step to DHCP server.



Here we used the copper straight-through cables to connect each component to your due place, the pc HR VLAN its in switch input fa0/2 and HR VLAN 2 in fa0/3, the IT pc is in the fa0/4, and i leave the fa0/1 to the server.



Now we have to creat Vlan’s to HR and IT.

That's the settings of the pc 1.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
```

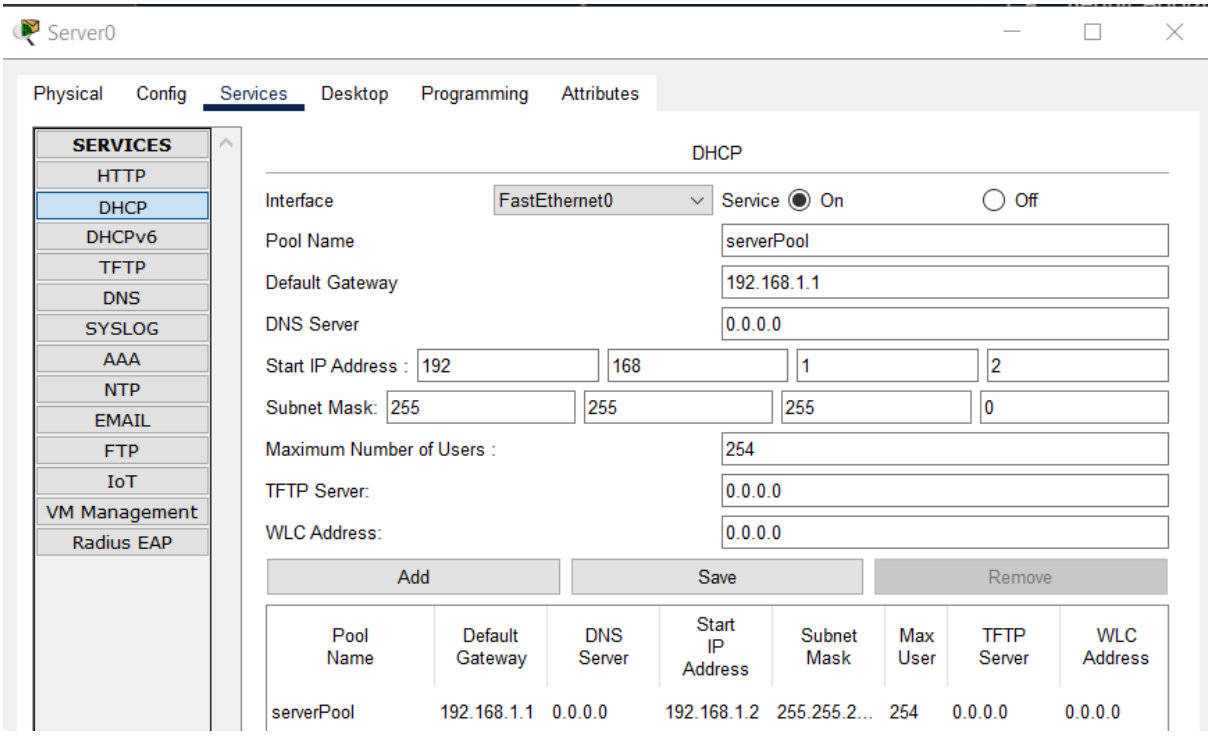
Here pc 2.

```
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
```

Here pc 3, note that it is in another vlan.

```
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
```

Here we have to configure the DHCP server so that it can assign automatic IPs to our PCs, we let the 192.168.1.1 IP to the server, so we gonna have from 192.168.1.2, that's the start,we let the maximum in 254, but if we want, we can change the assign for how many we wanted,so we can use 253 IP address in our local lan.



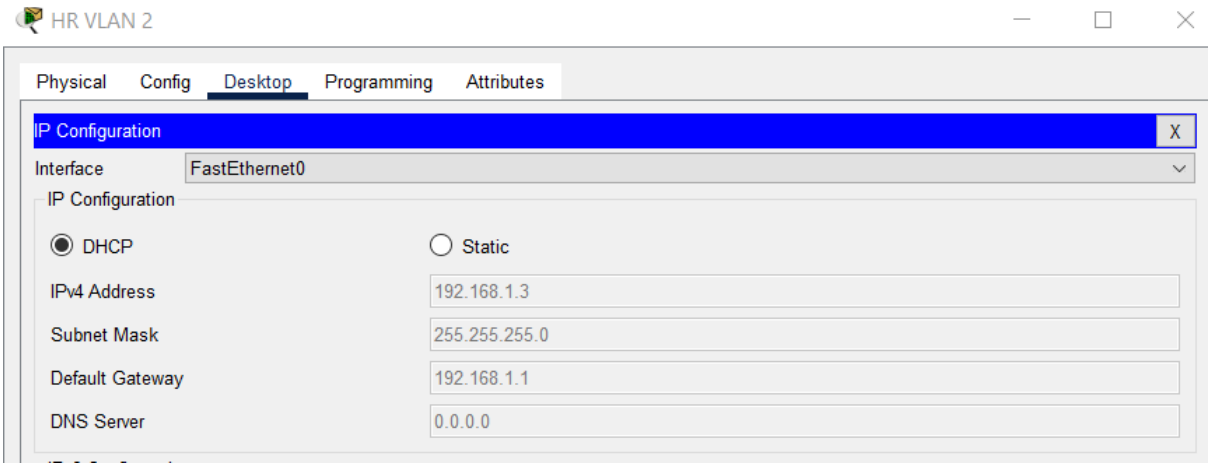
Now, lets try some connectivities, we gonna do some pings in our PCs,

Here I did two pings, one to the server and other to the pc 2, and notice that both is working, because they are on the same VLAN, all good, we don't lose any packet.

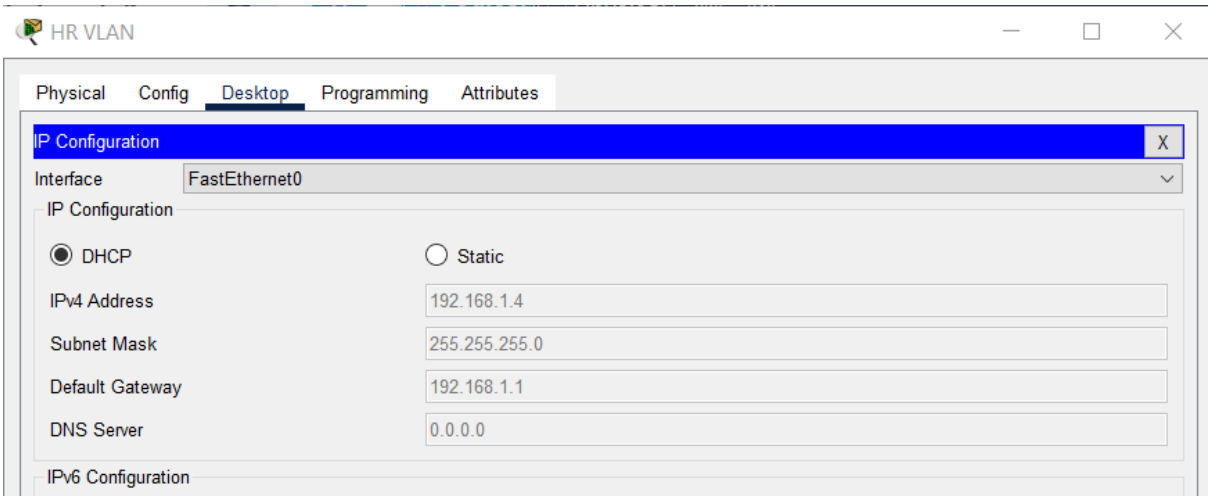
```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

Here we can see that the DHCP server its working, because we got DHCP IPs.

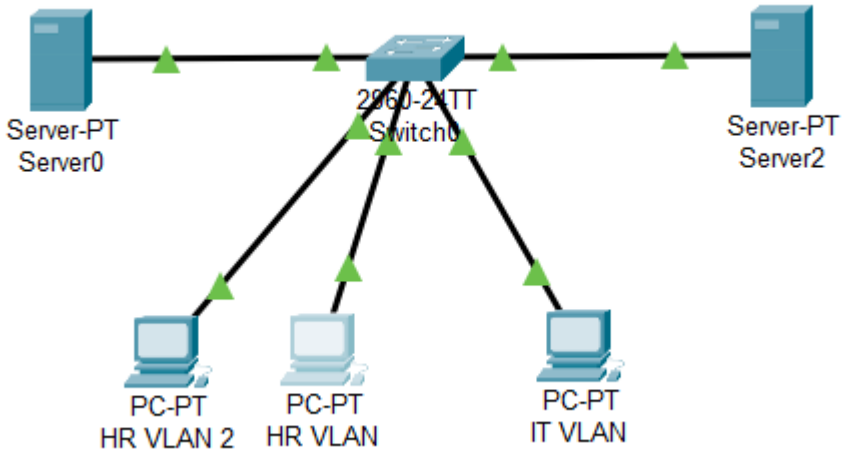
pc 1.



pc 2.

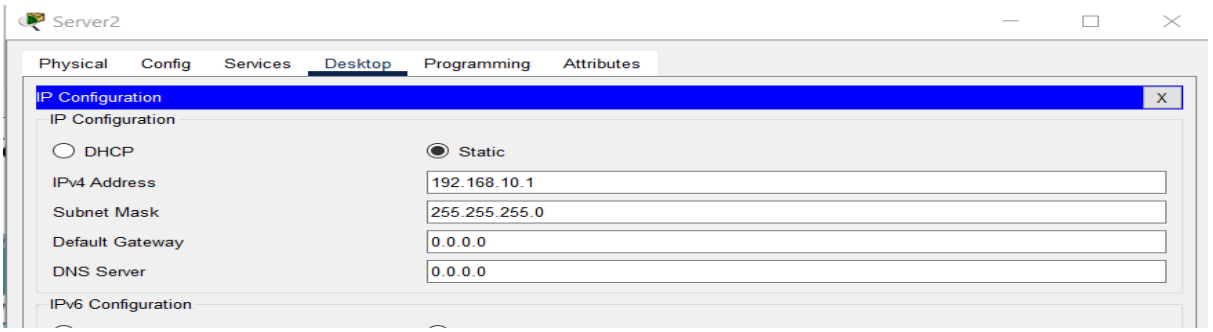


To get an IP for pc 3 in VLAN 20, we had to insert another server,as each server only works for each VLAN.

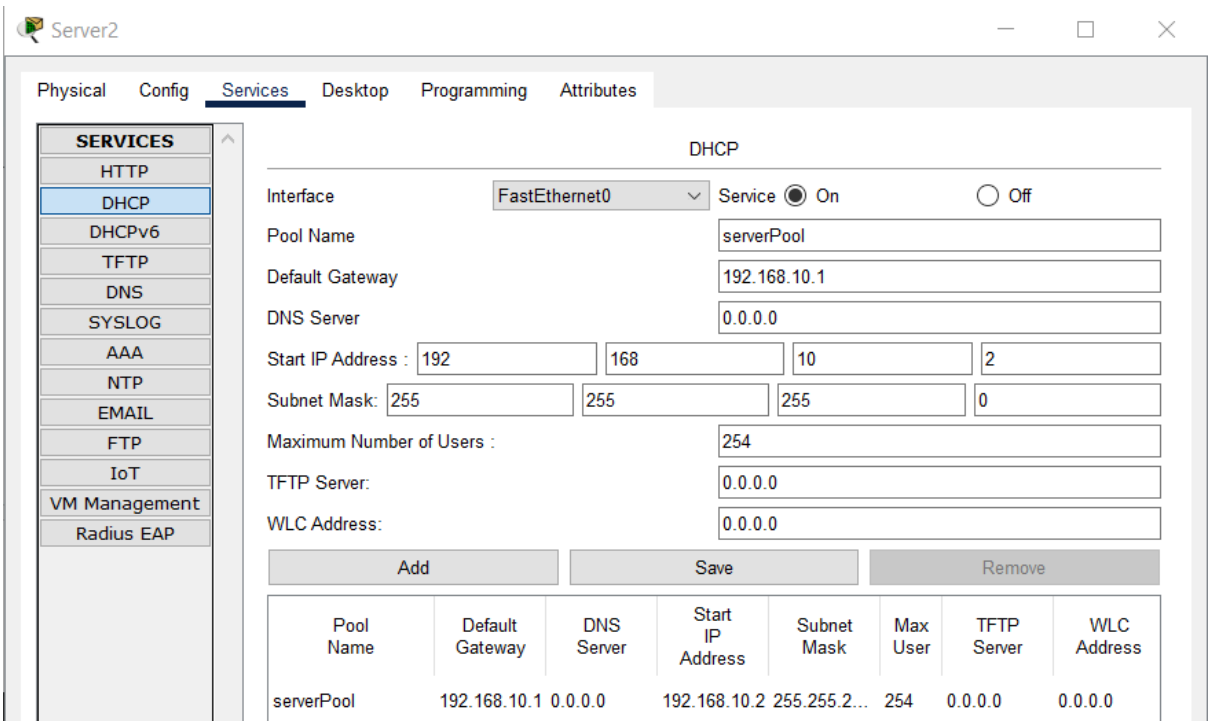


Lets see the settings about this new server.

That's the static IP, 192.168.10.1, notice that it has to be different to the other server that we have.



Thats our settings inside the server, so that can assign automatic IPs.



Now we tried to do two pings, one in the same vlan 10, i used the pc 2 to ping the pc 1, “marked in red”, notice that worked, but when i tried do a ping in pc 3 “marked in green” in the other server it does not work, because we have to stay in the same lan.

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Here a ping by pc 3 to the pc 1, just to see that does not work.

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Here a ping by pc 1 to the server, successful.

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Here a ping by pc 2 to the server, successful also.

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```